

Subject-Specific Lower Limb Waveforms Planning via Artificial Neural Network

Trieu Phat Luu, H. B. Lim, Xingda Qu, K. H. Hoon, and K. H. Low
School of Mechanical and Aerospace Engineering
Nanyang Technological University (NTU)
Singapore 639798

Abstract— Robotic is gaining its popularity in gait rehabilitation. Gait pattern planning is important, in order to ensure the gait patterns induced by robotic systems on the patient are natural and smooth. It is known that the gait parameters (stride length, cadence) are the key factors, which affect gait pattern. However, a systematic methodology for gait pattern planning is missing. Therefore, a gait pattern generation methodology, *GaitGen*, was proposed in our previous work. In this paper, we introduce a new model to enhance the proposed methodology for generating the joint angle waveforms of the lower limb during walking, with the gait parameters and the lower limb anthropometric data as input. The walking motion was captured with a motion capture system using passive markers. The waveforms of lower limb joint angles were calculated from the experimental data and the waveforms were then decomposed into Fourier coefficients. Therefore, each joint angle waveform can be represented by a Fourier coefficient vector containing eleven elements to facilitate the waveform analysis. Multi-layer perceptron neural networks (MLPNNs) were designed to predict the Fourier coefficient vectors for specific subject and desired gait parameters. Assessment parameters such as correlation coefficient, mean absolute deviation (MAD) and threshold absolute deviation (TAD) were calculated to examine the quality of MLPNNs' prediction. The constructed waveforms from predicted Fourier coefficient vectors were compared with the actual waveforms calculated from experimental data by using the above-mentioned assessment parameters. The results show that the constructed waveforms closely match the experimental waveforms based on the assessment parameter outcomes.

I. INTRODUCTION

To maximize the therapeutic outcome, it is crucial to induce a gait pattern that resembles natural human gait pattern during gait rehabilitation [1, 2]. In the context of practicing correct kinematic gait patterns in a repetitive manner, little works have been focused on the study of planning gait patterns for patients. In therapist assisted body weight supported (BWS) treadmill gait rehabilitation, the therapist moves the patient leg and pelvic based on their visual feedback and "feel". The assistance provided can vary greatly between therapists and between training sessions [3].

Robotic gait rehabilitation systems required a pre-defined pattern of gait locomotion, in order to carry out the gait training. Robotic orthosis found in gait rehabilitation systems adopt simplified approach to replicate the leg kinematics [4, 5]. Lokomat took a step further by allowing the gait pattern to be planned according to the patient's height and range of motion

of the lower limb joints [6]. However, most of these gait planning methods are template based, whereby motion templates of subjects with anatomical parameters similar to the targeted subjects are required.

Motivation

Template based gait planning approach has one limitation. The collection of template needs to cover a wide spectrum of subjects of different anatomical parameters and various walking speeds. It is not physically feasible to gather all necessary templates to match each targeted subject perfectly.

Motivated to provide a better gait planning approach for robotic gait rehabilitation systems, a systematic methodology for gait pattern generation, *GaitGen*, has been introduced [7]. Lower limb joint angle waveforms generation for intra-subject has been proposed in previous work [8].

This work introduces a new model to generate the lower limb joint angle waveforms in the sagittal plane for multiple subjects. In this work, the multi-layer perceptron neural networks (MLPNNs) are designed and trained for the estimation of the Fourier coefficient vectors for specific subjects with desired gait parameters. Lower limb joint angle waveforms are then generated from the estimated Fourier coefficient vectors.

Organization of Paper

The structure of this paper is organized as follows: Section II introduces gait planning methodology for rehabilitation. Section III gives the details on the design of gait experiment. Section IV presents the design of MLPNNs. Section V summarizes the key points discussed in this paper and suggests future works that will further enhance the current work.

II. METHODOLOGY OF GAIT PATTERN PLANNING

After neurologic injury or disease, patients are trained to restore or enhance walking ability through the process of locomotor training. One of the crucial sensory cues in locomotor training is to approximate normal hip, knee, and ankle kinematics for walking [2].

GaitGen methodology was designed to generate gait pattern for specified subject [7]. This methodology provides a systematic approach of gait pattern generation for robotic assisted gait rehabilitation. A gait pattern specifically tailored for a patient can be generated based on the patient's anatomical parameters.

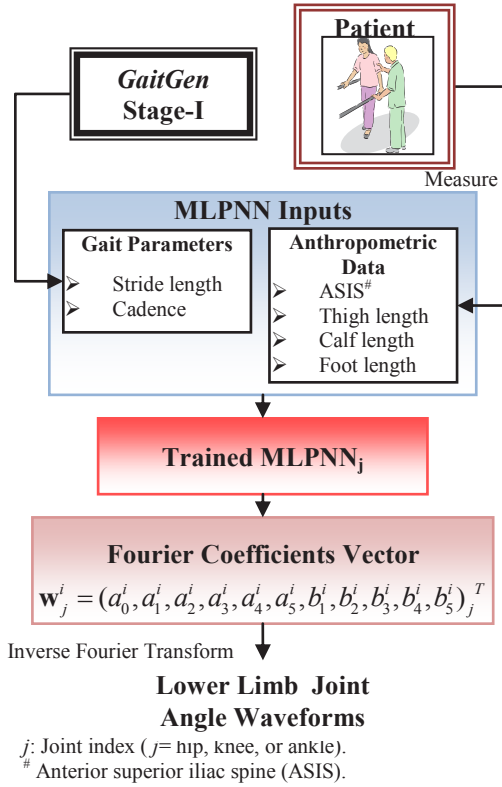


Figure 1. Overview of *GaitGen*

GaitGen was designed with two stages. Stage-I of *GaitGen* was designed to estimate natural walking speed, stride length, and cadence for a specific subject, given anthropometric data of the subject and desired walking speed state (slow or normal). The work on the Stage-I has been published in reference [9]. Stage-II of *GaitGen* uses the stride length and cadence estimated in Stage-I to generate a corresponding gait pattern.

Previous work on Stage-II of *GaitGen* proposed a multiple linear regression model to compute the Fourier coefficient vectors from given gait parameters. Subsequently, the lower limb joint angle waveforms are generated from the Fourier coefficient vectors [8]. This linear function is limited to intra-subject case. In this paper, the Stage-II of *GaitGen* is enhanced by introducing a new model for the computation of Fourier coefficient vectors, so that the Stage-II of *GaitGen* is applicable for inter-subjects case. Multi-Layer Perceptron Neural Network (MLPNN) is applied to estimate the Fourier coefficient vectors for specific subjects and desired gait parameters. The proposed Stage-II of *GaitGen* is shown in Fig.1

III. EXPERIMENTAL SETUP AND ANALYSIS METHOD

A. Experimental Setup

The data of gait motion for this work was collected at a frequency of 100 Hz from the Eagle digital motion capture system developed by Motion Capture Analysis Corporation [10]. It is a passive marker based motion capture system. Eight cameras were used to capture a work envelope for approximately three complete gait cycles. The walkway is about ten meters long, and capture volume is positioned at the

middle of the walkway. Adequate walkway space is provided to ensure the subject enters the capture volume at rhythmic walking stage. This work follows the markers set introduced by Christopher [11]. Positions of lower limb joint centers and segmental reference frames are calculated from the position data of markers and the lower limb joint angles are calculated based on these segmental reference frames [11].

Fifteen healthy male subjects with ages ranging 18 to 26 years (mean 22.3 ± 1.9 years), height 160- 180 cm (mean 172.2 ± 4.9 cm) and weight 49- 75 kg (mean 61.9 ± 8.5 kg), were recruited from the local student population of the Nanyang Technological University (NTU), Singapore. These subjects have neither neurological injury nor gait disorder. Each subject was instructed to walk ten times at their slow and self-paced walking speed. The general information of the subjects and number of walks are shown in Table I.

TABLE I. INFORMATION OF SUBJECTS AND NUMBER OF WALKS

Number of subjects	Gender	Age	Number of slow walks	Number of normal walks	Number of total walks
15	Male	18-26	10	10	300

B. Lower limb joint angle waveforms in frequency domain

The method of applying Fast Fourier Transform (FFT) to analyze the lower limb joint angle waveforms in frequency domain was introduced in previous work [8]. This work demonstrated that the lower limb joint angle waveforms are mainly contributed by signals which have frequency less than 6 Hz; therefore, each joint angle waveform can be compressed, represented and reconstructed by using an 11 elements Fourier coefficients vector as follow:

$$f(t_n) = \frac{1}{2}a_0 + \sum_{p=1}^5 \left[a_p \cos\left(\frac{2\pi pn}{N}\right) + b_p \sin\left(\frac{2\pi pn}{N}\right) \right] \quad (1)$$

$n = 0, 1, \dots, N-1$

where a_p and b_p is the Fourier coefficients, and N is the number of samples in one gait cycle collected from the motion capture experiment.

The Fourier coefficient vector is formed as

$$\mathbf{u}_j^i = (a_0^i, a_1^i, a_2^i, a_3^i, a_4^i, a_5^i, b_1^i, b_2^i, b_3^i, b_4^i, b_5^i)_j^T \quad (2)$$

whereby i is the number of walk ($i=1,2,\dots,300$) and j is the joint index (j = hip, knee, or ankle). For example, Fourier coefficient vector for hip joint of the walk-1 is:

$$\mathbf{u}_{hip}^1 = (a_0^1, a_1^1, a_2^1, a_3^1, a_4^1, a_5^1, b_1^1, b_2^1, b_3^1, b_4^1, b_5^1)_{hip}^T$$

The results in previous work [8] demonstrated that the reconstructed waveforms closely fit the waveforms calculated from the experimental data.

C. Limitations of multiple linear regression model in estimating the Fourier coefficient vector

In previous work [8], multiple linear regression model was proposed to compute the Fourier coefficient vectors from a

given set of gait parameters (stride length, cadence). When the Fourier coefficient vectors are computed, the lower limb joint angle waveforms can be constructed and a gait pattern is planned for a specified subject. The initial results showed that the reconstructed waveforms from calculated Fourier coefficient vectors are very close to the actual waveforms obtained via motion capture experiment [8].

However, the multiple linear regression model takes only gait parameters as input and it is only applicable for intra-subject. When inter-subject is taken into consideration, the accuracy of this model is reduced significantly. This is because even though the same gait parameters were observed from two different subjects, the lower limb joint angle waveforms were different. It can be verified by analyzing the lower limb joint waveforms of two subjects with similar gait parameters. The lower limb anthropometric data and gait parameters of the two subjects are presented in Table II.

TABLE II. LOWER LIMB ANTHROPOMETRIC DATA AND GAIT PARAMETERS OF SUBJECT A AND SUBJECT B

	Lower limb anthropometric data				Gait parameters	
	ASIS breadth (cm)	Thigh length (cm)	Shank length (cm)	Foot length (cm)	Stride length (cm)	Cadence (steps/minute)
Subject A	25.3	40.6	41.9	27.5	125.5	54.1
Subject B	23.2	51	45	25.8	125.3	55

Lower limb joint angle waveforms of two subjects with the similar stride length and cadence are compared. The potential influences of anthropometric data of the lower limb towards the lower limb joint angle waveforms are highlighted by comparing the waveforms of two subjects with similar gait parameters. Fig. 2 shows the hip, knee, ankle, and metatarsal trajectories during walking for the two subjects. Subject A walked at the speed of 114.9 cm/s (stride length is 125.3 cm and cadence is 55 steps/minute) and Subject B walked at speed of 113.2cm/s (stride length is 125.5 cm and cadence is 54.1 steps/minute). The walks of the two subjects are intentionally selected from our database obtained via motion capture experiment. A perfect match of these three gait parameters of two subjects is not physically possible. Thus, walks with similar gait parameters from two subjects are compared.

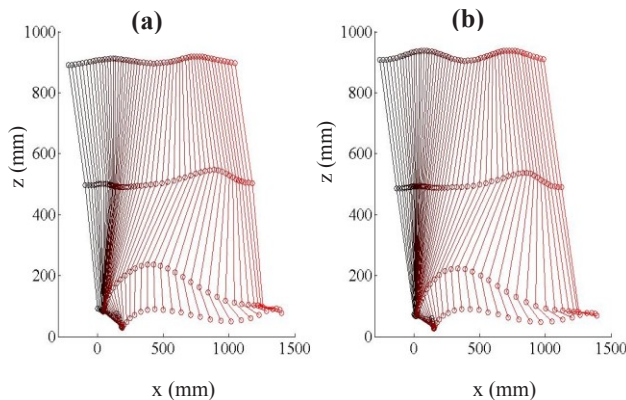


Figure 2. Lower limb joint trajectories in sagittal plane: (a) Subject A; (b) Subject B

Fig. 3 presents joint angles of hip, knee, and ankle in the sagittal plane for the walk of the two subjects. A clear difference is found between two set of lower limb joint angle waveforms, despite of they are walking at similar walking speed, with comparable stride length and cadence. Based on this finding, we can hypothesize that lower limb joint angle waveforms are influenced by anthropometric data. As such, in the following analysis, anthropometric data will be included as one of the inputs and MLPNNs are proposed as a non-linear model to estimate the Fourier coefficient vectors.

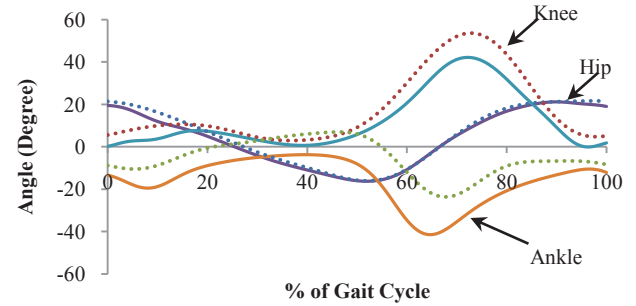


Figure 3. Lower limb joint angle waveforms (Solid lines- subject A, dotted lines- subject B)

IV. DESIGN OF MLPNN

Neural network is a prevalent analysis tool used in gait data studies in the last 10 years [12]. Neural network provides gait data analysis with a highly flexible, non-linear modeling ability. The study of complicated gait variable relationships which have been difficult to model analytically can be facilitated by the non-linear property of neural network. In this work, multi-layer perceptron neural networks (MLPNNs) are applied to calculate the Fourier coefficient vectors when the gait parameters (stride length, cadence) and the lower limb anthropometric data of the subject are provided as input vector.

Selection of MLPNN inputs

As mentioned in section III, the lower limb joint angle waveforms are not only affected by gait parameters but also the lower limb anthropometric data. Based on literature review and analysis of gait kinematics, we hypothesized that these factors: stride length, cadence, ASIS, thigh length, shank length and foot length significantly influence the lower limb joint angle waveforms. Gait parameters (stride length, cadence) are obtained by using stage-I of *GaitGen* while the lower limb anthropometric data (ASIS, thigh length, shank length and foot length) can be obtained by measurement.

MLPNN architecture design

The proposed MLPNNs are designed to have one input layer, one hidden layer and one output layer. The input layer contains 6 neurons for input vector $\mathbf{v}_j^l$ formed by the gait parameters and lower limb anthropometric data. The output layer contains 11 neurons for Fourier coefficient vectors $\mathbf{w}_j^l$ (j = hip, knee or ankle). The optimized number of neurons in the hidden layer is denoted as K_j and it is determined by trial and error method, $b_{j,k}^l$ is the bias value of neuron k of layer l . For example, $b_{hip,1}^1$ is the bias value of the first neuron in the first

layer of the MLPNN_{Hip}. The MLPNNs' architectures are depicted in Fig. 4

Each neuron in the MLPNNs uses the hyperbolic tangent sigmoid transfer function as the activation function. The sigmoid function is derivative and provides the output in the range of [-1, 1]. The activation function is given by

$$\varphi_k[\gamma_k(q)] = c \frac{e^{d\gamma_k(q)} - e^{-d\gamma_k(q)}}{e^{d\gamma_k(q)} + e^{-d\gamma_k(q)}} \quad (3)$$

where

- φ_k : Activation function of neuron k
- γ_k : Activation signal of neuron k
- c, d : Constant value.

Lavenberg-Marquardt (LM) algorithm is used as learning algorithm in the MLPNNs. This algorithm is currently considered as the most efficient when the network contains no more than a few hundred weights [13, 14]

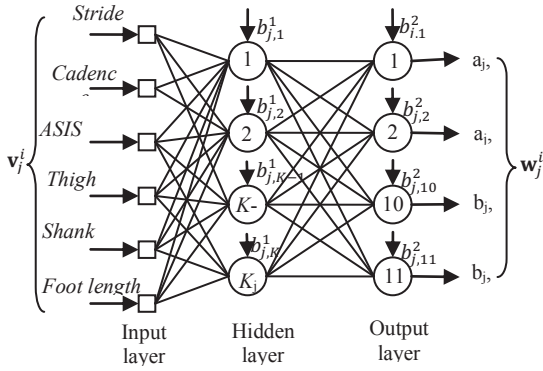


Figure 4. Architecture of trained MLPNN, j is the joint index (hip, knee, ankle) and K is the number of neurons in the hidden layer

The proposed MLPNNs are trained with input vectors $\mathbf{r}_j^i$ and output vectors $\mathbf{u}_j^i$ from the training set. The division of experimental data is shown in Table III. These vectors are derived from the experimental data. In order to obtain the optimized number of neurons required in the hidden layers, the MLPNNs are trained and tested with a variation of 5 to 70 neurons in the hidden layers. The output vectors $\mathbf{w}_j^i$ are obtained from the input vector $\mathbf{v}_j^i$ in the test set. The output vectors $\mathbf{w}_j^i$ are then compared to the actual Fourier coefficient vectors to determine the coefficient of determination (or R-squared) of the MLPNN models.

The R-squared value of each MLPNN versus number of neurons in the hidden layer is depicted in Fig. 5. It is noticeable that when the number of neurons in the hidden layer is not so big, the R-squared value increases with the number of neurons. However, if the number of neurons is large, the R-squared value decreases and becomes unstable. It is because the neural network will become over-trained, unstable and its generalization capacity is also lower when the number of neurons in the hidden layer is large [15]. The number of neurons which provides stable, high value of R-squared will be adopted into the neural network. Based on the result,

MLPNN_{Hip}, MLPNN_{Knee} and MLPNN_{Ankle} are designed to have 13, 18 and 15 neurons in the hidden layers, respectively.

TABLE III. DIVISION OF EXPERIMENTAL DATA FOR MLPNN

	Number of walks	Input vector	Output vector
Training set	250	$\mathbf{r}_j^i$	$\mathbf{u}_j^i$
Test set	50	$\mathbf{v}_j^i$	$\mathbf{w}_j^i$

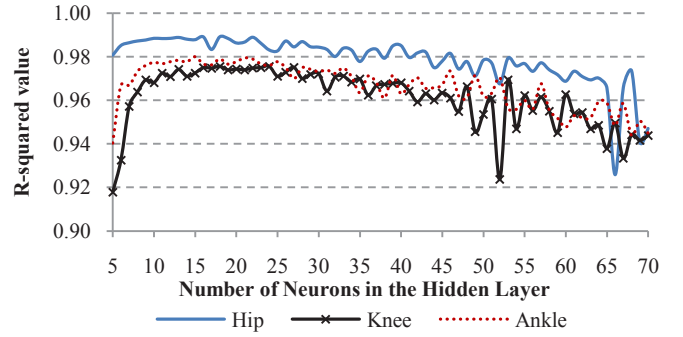


Figure 5. The R-squared values of MLPNNs for different number of neurons in the hidden layers.

V. RESULTS AND DISCUSSION

A. Assessment parameters for evaluating quality of MLPNNs' estimation.

By applying the Inverse Fourier Transform shown in (1), the lower limb joint angle waveforms can be planned for a specified subject from Fourier coefficient vectors estimated by MLPNNs. In order to assess the quality of MLPNNs' estimation, comparison is made to examine the difference between the estimated angle waveforms and the actual angle waveforms calculated from experimental data. The quality of MLPNN models can be evaluated by using the following quantitative parameters:

- Correlation coefficient

$$\rho_j = \frac{\sigma_{\hat{\theta}_j \theta_j}}{\sigma_{\hat{\theta}_j} \sigma_{\theta_j}} \quad (4)$$

whereby j is the joint index, θ_j are the actual angles of joint j measured from the reflective markers, $\hat{\theta}_j$ are the estimated angles, $\sigma_{\hat{\theta}_j \theta_j}$ is the covariance and $\sigma_{\hat{\theta}_j}^2$, $\sigma_{\theta_j}^2$ is the variance.

- Mean Absolute Deviation (MAD)

$$MAD_j = \sum_{n=0}^{N-1} \frac{|\hat{\theta}_{j,n} - \theta_{j,n}|}{N} \quad (5)$$

whereby N is the number of samples in one gait cycle collected from the motion capture experiment, MAD is measured in degree and it determines how much the estimated angles are close to the actual angles.

- The third parameter used to measure the quality of MLPNNs' estimation is the Threshold Absolute Deviation (TAD). TAD determines the percentage of points which exceeds a pre-specified threshold around

the actual waveform. This parameter was introduced in [16], and the same 5° threshold is used in this work.

$$TAD_j = \frac{100}{N} \sum_{n=0}^{N-1} T_{j,n} \quad (6)$$

$$T_{j,n} = \begin{cases} 1, & \text{when } |\hat{\theta}_{j,n} - \theta_{j,n}| > \gamma \\ 0, & \text{otherwise} \end{cases}$$

B. Validation of reconstructed lower limb joint angle waveforms

MLPNNs are implemented to estimate the Fourier coefficient output vectors $\mathbf{w}_j^i$ for the test set (50 walks). The lower limb joint angle waveforms are reconstructed from $\mathbf{w}_j^i$ and the assessment parameters are calculated to measure the quality of MLPNNs estimation. The mean and standard deviation of these parameters are calculated and shown in Table IV. The results show that the correlation coefficient between estimated lower limb waveforms and the actual waveforms are very high (0.98-0.99), average value of MAD are low (1.27- 2.12°) and the percentage of errors exceeded the 5° threshold is small (0.95-9.07%).

Among the three MLPNNs, MLPNN_{Hip} has the best result of assessment parameter values and it contains the least number of neurons in the hidden layer (13 neurons). The reason being the hip joint angle waveform is relatively less complex than the knee and ankle joints waveforms.

TABLE IV. MEAN (STD) FROM 50 WALKS OF TEST SET

	ρ	MAD (Degree)	TAD (%)
Hip	0.99(0.01)	1.27(0.53)	0.95(4.08)
Knee	0.98(0.02)	2.12(1.26)	9.07(14.01)
Ankle	0.98(0.02)	1.55(0.61)	5.11(7.05)

C. Example results of two walks from test set

Two walks in the test set are used as examples to illustrate the quality of the trained MLPNNs. These two walks are randomly selected from our database for different subjects with different gait parameters (stride length, cadence). The information of these two walks is shown in Table V.

TABLE V. INFORMATION OF TWO WALKS FROM TEST SET

	Lower limb anthropometric data				Gait parameters	
	ASIS breadth (cm)	Thigh length (cm)	Shank length (cm)	Foot length (cm)	Stride length (cm)	Cadence (steps/minute)
Subject 1	24.3	40.8	39.1	26.1	124.8	61.2
Subject 2	25.4	36.6	37.5	24.7	108.1	40.8

The actual Fourier coefficient vectors are calculated from experimental data for walk-1 and walk-2 ($\mathbf{u}_{hip}^1, \mathbf{u}_{knee}^1, \mathbf{u}_{ankle}^1, \mathbf{u}_{hip}^2, \mathbf{u}_{knee}^2, \mathbf{u}_{ankle}^2$) and the estimated Fourier coefficient vectors are provided from MLPNNs ($\mathbf{w}_{hip}^1, \mathbf{w}_{knee}^1, \mathbf{w}_{ankle}^1, \mathbf{w}_{hip}^2, \mathbf{w}_{knee}^2, \mathbf{w}_{ankle}^2$). Subsequently, the angle waveforms of hip, knee and ankle are reconstructed from the estimated Fourier coefficient

vectors by using (1). The actual waveforms obtained via motion capture experiment are presented together with the reconstructed waveforms in Figs. 6 and 7.

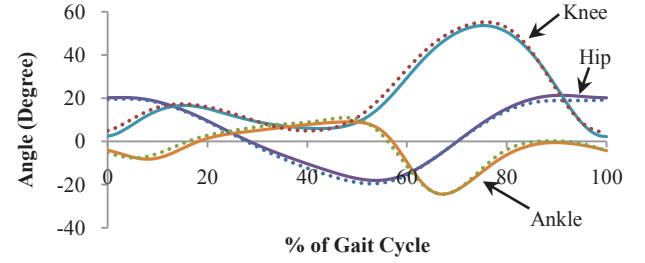


Figure 6. Lower limb joint angle waveforms for walk-1 (Solid lines-actual waveforms from experimental data, dotted lines- reconstructed waveforms using estimated Fourier coefficient vectors $\mathbf{w}_{hip}^1, \mathbf{w}_{knee}^1, \mathbf{w}_{ankle}^1$)

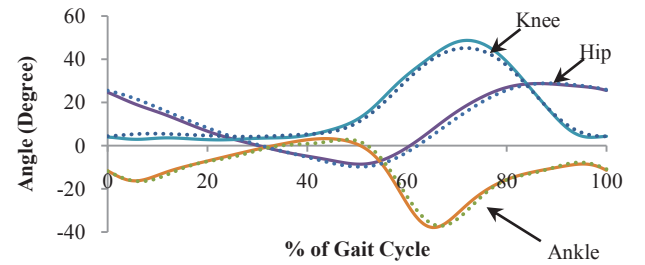


Figure 7. Lower limb joint angle waveforms for walk-2 (Solid lines-actual waveforms from experimental data, dotted lines- reconstructed waveforms using estimated Fourier coefficient vectors $\mathbf{w}_{hip}^2, \mathbf{w}_{knee}^2, \mathbf{w}_{ankle}^2$)

The assessment parameters are calculated to examine the deviation of estimated angle waveforms from the actual waveforms. The results of the examination are shown in Table VI. The mean absolute deviations are less than 2° for all of the angle waveforms and the angles are not exceeding the threshold. The assessment parameters confirm that the estimated lower limb joint angle waveforms are very close to the actual waveforms.

The results show that the proposed method is capable of generating the waveforms for a wide range of walking speed. In the previous examples of the two walks, subject 1 and subject 2 walks at speed of 127.3 and 73.5 cm/s, respectively. The estimated angle waveforms closely matched the actual waveforms for both walks (Figs. 6 and 7).

TABLE VI. ASSESSMENT PARAMETERS FOR ANGLE WAVEFORMS OF WALK-1 AND WALK-2

	ρ		MAD (Degree)		TAD (%)	
	Walk-1	Walk-2	Walk-1	Walk-2	Walk-1	Walk-2
Hip	1	1	1.95	0.79	0	0
Knee	1	1	1.77	1.21	0	0
Ankle	1	1	1.06	0.8	0	0

VI. CONCLUDING REMARKS

In this work, the multi-layer perceptron neural networks (MLPNNs) are proposed for the estimation of the Fourier coefficient vectors, which are used for lower limb joint angle waveforms construction. After MLPNNs are trained, these models will estimate the value of Fourier coefficient vectors when the anthropometric data and gait parameters of a specific subject are given. Using the estimated Fourier coefficient vectors, the lower limb joint angle waveforms can be constructed for the subject by applying (1). The constructed waveforms from estimated Fourier coefficients closely matched the actual waveforms calculated from experimental data.

In conclusion, this paper has presented a systematic methodology for subject tailored lower limb joint angle waveforms planning. The method shows the ability of generating lower limb joint angle waveforms for a specific subject when their anthropometric data and gait parameters are given. This method will be used for planning a gait pattern for specific patient with a wide range of walking speed, since the motion capture experiment cannot be used to obtain a useful gait pattern from the patient for gait rehabilitation training process. This method can also be applied in patient cooperative control strategy to update the gait pattern quickly during the training process based on the performance of the patient [17]. For example, when the patient tends to increase the stride length, the Stage-II of *GaitGen* will estimate the new Fourier coefficient vectors and update the gait pattern for the robotic gait rehabilitation system.

The proposed methodology and MLPNNs have shown promising results using a small database for training the MLPNNs. It is therefore essential in future work to enlarge the training data set to cover a wider range of anthropometric parameters for male and female subjects to enhance the accuracy of the proposed methodology.

ACKNOWLEDGMENTS

The authors would like to thank the Department of the Rehabilitation Medicine at the Tan Tock Seng Hospital (TTSH), the equipment and financial supports provided by the NMRC Research Grant 1051/2006 and the Robotics Research Center (RRC) of the Nanyang Technological University (NTU) for their support to this work. The authors would also like to acknowledge Mr. Narendra Utama for his assistance in the data collection.

REFERENCES

- [1] H. Barbeau and S. Rossignol, "Recovery of locomotion after chronic spinalization in the adult cat," *Brain Research*, vol. 412, pp. 84-95, 1987.
- [2] A. L. Behrman and S. J. Harkema, "Locomotor training after human spinal cord injury: A series of case studies," *Physical Therapy*, vol. 80, pp. 688-700, 2000.
- [3] J. A. Galvez, G. Kerdanyan, S. Maneekobkunwong, R. Weber, M. Scott, S. J. Harkema, and D. J. Reinkensmeyer, "Measuring human trainers' skill for the design of better robot control algorithms for gait training after spinal cord injury," in *Proceedings of the 2005 IEEE 9th International Conference on Rehabilitation Robotics*, Chicago, IL, vol. 2005, pp. 231-234, 2005.
- [4] G. Colombo, M. Joerg, R. Schreier, and V. Dietz, "Treadmill training of paraplegic patients using a robotic orthosis," *Journal of Rehabilitation Research and Development*, vol. 37, pp. 693-700, 2000.
- [5] C. Y. E. Wang, J. E. Bobrow, and D. J. Reinkensmeyer, "Dynamic motion planning for the design of robotic gait rehabilitation," *Journal of Biomechanical Engineering*, vol. 127, pp. 672-679, 2005.
- [6] R. Rienen, L. Lünenburger, I. Maier, G. Colombo, and V. Dietz, "Locomotor Training in Subjects with Sensori-Motor Deficits: An Overview of the Robotic Gait Orthosis Lokomat," *Journal of Healthcare Engineering*, vol. 1, pp. 197-216, 2010.
- [7] H. B. Lim, K. H. Hoon, Y. C. Soh, Adela Tow, and K. H. Low, "Gait planning for effective rehabilitation - from gait study to application in clinical rehabilitation," in *Proceedings of the 2009 IEEE International Conference on Rehabilitation Robotics, ICORR 2009*, Kyoto, Japan, pp. 271-276, 2009.
- [8] Trieu Phat Luu, H. B. Lim, Xingda Qu, and K. H. Low, "Subject Tailored Gait Pattern Planning for Robotic Gait Rehabilitation," in *Proceedings of the International Conference on Robotics and Biomimetics (ROBIO2010)*, Tianjin, China, Dec 2010.
- [9] H. B. Lim, Trieu Phat Luu, K. H. Hoon, and K. H. Low, "Natural Gait Parameters Prediction for Gait Rehabilitation via Artificial Neural Network," *Accepted for Presentation at 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Taipei, Taiwan, 2010.
- [10] "Motion Analysis Corporation, the Motion Capture Leader," 2009. [Online]. Available: <http://www.motionanalysis.com/>. Accessed on March, 2010
- [11] C. L. Vaughan, B. L. Davis, and J. C. O'Connor, *Dynamics of Human Gait*. Kiboho Publishers, 1999.
- [12] T. Chau, "A review of analytical techniques for gait data. Part 2: Neural network and wavelet methods," *Gait and Posture*, vol. 13, pp. 102-120, 2001.
- [13] B. M. Wilamowski, Y. Chen, and A. Malinowski, "Efficient algorithm for training neural networks with one hidden layer," in *Proceedings of the International Joint Conference on Neural Networks*, Washington, DC, USA, vol. 3, pp. 1725-1728, 1999.
- [14] M. T. Hagan and M. B. Menhaj, "Training feedforward networks with the Marquardt algorithm," *Neural Networks, IEEE Transactions on*, vol. 5, pp. 989-993, 1994.
- [15] K. Shin-ike, "A two phase method for determining the number of neurons in the hidden layer of a 3-layer neural network," in *SICE Annual Conference 2010, Proceedings of*, pp. 238-242, 18-21 Aug. 2010.
- [16] A. Findlow, J. Y. Goulermas, C. Nester, D. Howard, and L. P. J. Kenney, "Predicting lower limb joint kinematics using wearable motion sensors," *Gait and Posture*, vol. 28, pp. 120-126, 2008.
- [17] S. Jezernik, G. Colombo, and M. Morari, "Automatic gait-pattern adaptation algorithms for rehabilitation with a 4-DOF robotic orthosis," *IEEE Transactions on Robotics and Automation*, vol. 20, pp. 574-582, 2004.